UNIVERSITI TEKNOLOGI MARA

GROUP PROJECT

COURSE	**:**	**DIGITAL SIGNAL PROCESSING**
COURSE CODE	**:**	**ESE563**
SEMESTER	**:**	**MARCH 2025**

"I hereby declare that all submission is based on my original efforts, not plagiarised from any source(s), nor attempting to or communicating with any other person. I am fully aware that candidates who are suspected of violating this pledge are liable to be referred to the Disciplinary Committee of the College."

NAME & UiTM NO. :

1) **SYAZA NABILA BINTI ZAHARIN (2023836486)**
2) **ANIS SUHANA BINTI MOHD AZMI (2023816852)**
3) **SYIFAA' NURAIMUNI BINTI SYAHRUL NIZAM (2023801088)**

GROUP : CEEE222T6A

LECTURER : DR. MEGAT SYAHIRUL AMIN MEGAT ALI

Task	CO	PO	Level of Difficulty	WP, WKs, SDG	MARKS EARNED
1	CO3	PO3	C2		/5
2	CO3	PO3	C6		/20
3	CO3	PO3	C6		/10
4	CO3	PO3	C5		/5
5	CO3	PO3	C6		/10
				MARKS	/50

Task 1 (5 marks)

Explain the problem and its consequences. This is to show the importance of your proposed method.

1. Explain the problem that you are solving and its consequences.

The electrical activity of the brain is recorded by electroencephalogram (EEG) signals, which include significant frequency components like beta, theta, alpha, and delta waves. EEG data are frequently tainted by noise, interference, and frequencies from different brainwave bands that overlap, though.

Alpha waves, which are essential for examining states of calm, relaxation, and meditation, are what we hope to isolate in this project. If these alpha wave signals are not properly filtered, they may be obscured or distorted by noise and undesired components (such as beta or delta waves), which could result in an inaccurate assessment of the brain's health. This might lead to poor performance in brain-computer interface systems, incorrect diagnoses, or misleading findings in neuroscience studies.

A window technique Finite Impulse Response (FIR) filter is suggested as a solution to this problem. It efficiently extracts the alpha wave frequency band from the EEG data, guaranteeing that the signal is clear and focused on the desired frequency range.

2. Determine the frequency range for alpha waves.

The usual frequency range for alpha waves is 8 Hz to 13 Hz.
While attenuating other frequency bands, the filter should maintain this desired frequency band.

3. Assess the type of filter that is suitable for this application.

The best kind of filter to use since we want to suppress frequencies outside of the 8–13 Hz range and preserve only that range is a:

The Bandpass Filter

While attenuating both higher frequencies (such as beta and noise) and lower frequencies (such as delta and theta), a bandpass FIR filter allows the alpha wave frequencies to pass through.

TASK 2 (20 marks)

Formulate the proposed digital FIR filter design. Explain the concept of method that you have proposed.

4. Explain and formulate the impulse response for your filter.

There are two main types of digital filters: Finite Impulse Response (FIR) and Infinite Impulse Response (IIR) filters (de Cheveigné & Nelken, 2019). In this mini project, we focus on FIR filters, which produce a finite-duration output in response to an impulse input due to the absence of feedback, making them inherently stable. FIR filters operate by summing a finite number of weighted input samples. Mathematically, this is represented as:

$$y(n) = b_0 x(n) + b_1 x(n-1) + b_2 x(n-2) + .. + b_M x(n-M) \quad \text{...(1)}$$

or

$$y(n) = \sum_{k=0}^{M} b_k x(n-k) \quad \text{...(2)}$$

Where Bk is the feed forward filter coefficients, M is the length of the filter and k is the order of the filter. Using equation 2's z-transform, we will have:

$$Y(z) = \sum_{k=0}^{M} b_k z^{-k} X(z) \quad \text{...(3)}$$

In the more classical form as:

$$Y(z) = \mathrm{H(z).X(z)} \quad \text{...(4)}$$

where:

$$H(z) = \sum_{k=0}^{M} b_k z^{-k} \quad \text{...(5)}$$

This corresponds to the Z-transform of the impulse response and is useful for analyzing the frequency behavior of the FIR filter. The impulse response was formulated using the bandpass window method with a Hamming window.

For this project, FIR filter was designed to meet the following specifications:

Passband ripple:	$0.98 \leq \lvert H(e^{jw}) \rvert \leq 1.02$
Stopband ripple:	$\lvert H(e^{jw}) \rvert \leq 0.00224$
Transition width, Δw:	0.048π

Both lower cut-off frequency and upper cut-off frequency are normalized against f_s and converted into radial frequency used in the impulse response:

$$w_1 = 2\pi \frac{f_1}{f_s} = 2\pi \frac{8}{128} = 0.125\pi \; rad/s \quad \text{...(6)}$$

$$w_2 = 2\pi \frac{f_2}{f_s} = 2\pi \frac{13}{128} = 0.203\pi \; rad/s$$

Stopband attenuation A_s is then determined to select a suitable window function.

$$A_s = 20\log(0.00224) = -53dB \quad \text{...(7)}$$

Window method is functional to truncate an infinite ideal filter for this design. Hamming window was chosen due to its good trade-off between mainlobe width and sidelobe attenuation. Based on the specifications above it shows the maximum variation in the passband is ±2% from the ideal gain which is 1. In linear scale, it is written as ±0.02 or ±0.17dB in log scale. Meanwhile the value of 0.00224 shows the maximum gain allowed in the stopband which is -53dB attenuation. This controls how much unwanted frequencies leak into the output.

Since it is required for the filter to not exceed order of N=299 thus the alpha can be calculated:

$$\alpha = \frac{229}{2} = 114.5 \quad \text{...(8)}$$

For the proposed design, the filter combines both desired bandpass impulse response, $h_d(n)$ and hamming window w(n)

The desired impulse response for a bandpass filter,

$$h_d(n) = \frac{\sin[(0.203\pi(n - 114.5)] - \sin[0.125\pi(n - 114.5)]}{\pi(n - 114.5)} \quad \text{...(9)}$$

Furthermore, for window function,

$$w(n) = 0.54 - 0.46\cos\left(\frac{2\pi n}{N}\right), N = 229 \quad \text{...(10)}$$

The system FIR impulse response, h(n) is then obtained as,

$$h(n) = h_d(n) \, . \, w(n) \quad \text{...(11)}$$

$$h(n) = \left\{\frac{\sin[(0.203\pi(n-114.5)] - \sin[0.125\pi(n-114.5)]}{\pi(n-114.5)}\right\}. \left\{0.54 - 0.46\cos\left(\frac{2\pi n}{N}\right)\right\} \quad \text{...(12)}$$

At n=114.5, which is the center point, the peak amplitude of the desired bandpass filter's impulse response can be calculated as follows:

$$h(114.5) = \left\{\frac{0.2031\pi - 0.125\pi}{\pi}\right\}. 1 = 0.0781 \quad \text{...(13)}$$

At the center of the filter, where h=α=114.5, the ideal impulse response becomes an indeterminate form. Using the limit, the value is computed as $h_d(\alpha) = \frac{w_2 - w_1}{\pi} = 0.0781$. This value represents the normalized bandwidth of the filter and corresponds to the peak coefficient of the ideal bandpass response before applying the Hamming window.

The design and implementation of a bandpass FIR filter to separate alpha waves (8–13 Hz) from EEG data is the main objective of this paper. The window approach using a Hamming window was used to determine the filter's impulse response, h(n). Although additional EEG bands like theta (4–8 Hz) and beta (13–30 Hz) are also clinically significant, they are not included here for brevity because their formulation process is the same as that of the alpha filter.

5. Using Microsoft Excel, compute the filter coefficients.

N		229
fs		128
f1		0.0625
f2		0.1015625
α		114.5
w1 = 2πf1/fs		0.392699082
w2 = 2πf2/fs		0.638136008

The table above shows the basic parameters used in the design of an FIR filter to filter EEG alpha waves using the window method. The filter is designed with a length of N = 229, and the sampling frequency is 128 Hz. The alpha frequency range to be filtered is determined

by f1 = 0.0625 and f2 = 0.1015625, which are the normalization results from the range 8–13 Hz to units per Hz sampling. The alpha value of N/2 = 114.5 represents the center of the impulse response that is important in the calculation. Two corner values, w1 = 0.3927 and w2 = 0.6381, are used in the formulation of the ideal impulse function hD(n) for this band-pass filter. These values form the basis for calculating the filter coefficients and producing the desired frequency response.

n	n-α	hD(n)	w(n)	h(n)
0	-114.5	-0.004324891	0.08	-0.000345991
1	-113.5	-0.002037554	0.080173137	-0.000163357
2	-112.5	0.000720147	0.080692417	5.81104E-05
3	-111.5	0.003106918	0.081557449	0.000253392
4	-110.5	0.004438616	0.082767583	0.000367374
5	-109.5	0.004421485	0.084321907	0.000372828
6	-108.5	0.003236481	0.086219251	0.000279047
7	-107.5	0.001444644	0.088458187	0.000127791
8	-106.5	-0.000247296	0.091037029	-2.25131E-05
9	-105.5	-0.001267603	0.093953837	-0.000119096
10	-104.5	-0.001395332	0.097206414	-0.000135635
11	-103.5	-0.000826237	0.100792312	-8.32783E-05
12	-102.5	-6.38875E-05	0.104708832	-6.68958E-06
13	-101.5	0.000318383	0.108953026	3.46887E-05
14	-100.5	-5.38726E-05	0.113521699	-6.11571E-06
15	-99.5	-0.001169441	0.11841141	-0.000138475
16	-98.5	-0.002607648	0.123618481	-0.000322353
17	-97.5	-0.00369363	0.12913899	-0.000476992
18	-96.5	-0.003775579	0.134968783	-0.000509585
19	-95.5	-0.002514608	0.141103471	-0.00035482
20	-94.5	-6.81687E-05	0.147538435	-1.00575E-05
21	-93.5	0.002914189	0.154268832	0.000449569
22	-92.5	0.005492194	0.161289596	0.000885834
23	-91.5	0.006764733	0.168595441	0.001140503
24	-90.5	0.006206031	0.176180867	0.001093384
25	-89.5	0.003878643	0.184040165	0.000713826
26	-88.5	0.000437094	0.192167419	8.39953E-05
27	-87.5	-0.003085835	0.20055651	-0.000618884
28	-86.5	-0.005634903	0.209201123	-0.001178828
29	-85.5	-0.006499029	0.218094752	-0.001417404
30	-84.5	-0.005551337	0.227230701	-0.001261434
31	-83.5	-0.003272205	0.236602092	-0.000774211
32	-82.5	-0.000547605	0.246201872	-0.000134821
33	-81.5	0.001678284	0.256022814	0.000429679
34	-80.5	0.002760534	0.266057525	0.000734461
35	-79.5	0.00258132	0.276298452	0.000713215
36	-78.5	0.00155955	0.286737884	0.000447182
37	-77.5	0.000440387	0.297367965	0.000130957
38	-76.5	-4.86588E-05	0.308180691	-1.49957E-05
39	-75.5	0.000466707	0.319167924	0.000148958
40	-74.5	0.00182111	0.330321392	0.000601552

..... n; 0 - 229

210	95.5	-0.002514608	0.141103471	-0.00035482
211	96.5	-0.003775579	0.134968783	-0.000509585
212	97.5	-0.00369363	0.12913899	-0.000476992
213	98.5	-0.002607648	0.123618481	-0.000322353
214	99.5	-0.001169441	0.11841141	-0.000138475
215	100.5	-5.38726E-05	0.113521699	-6.11571E-06
216	101.5	0.000318383	0.108953026	3.46887E-05
217	102.5	-6.38875E-05	0.104708832	-6.68958E-06
218	103.5	-0.000826237	0.100792312	-8.32783E-05
219	104.5	-0.001395332	0.097206414	-0.000135635
220	105.5	-0.001267603	0.093953837	-0.000119096
221	106.5	-0.000247296	0.091037029	-2.25131E-05
222	107.5	0.001444644	0.088458187	0.000127791
223	108.5	0.003236481	0.086219251	0.000279047
224	109.5	0.004421485	0.084321907	0.000372828
225	110.5	0.004438616	0.082767583	0.000367374
226	111.5	0.003106918	0.081557449	0.000253392
227	112.5	0.000720147	0.080692417	5.81104E-05
228	113.5	-0.002037554	0.080173137	-0.000163357
229	114.5	-0.004324891	0.08	-0.000345991

From the table above that shows the computation the filter coefficients using Microsoft Excel. In the design of FIR (Finite Impulse Response) filters, the number of filter coefficients (or filter length) determines the accuracy and performance of the filtering. Since the maximum allowed order is 229, the number of samples is:

$$N+1=229+1=230$$

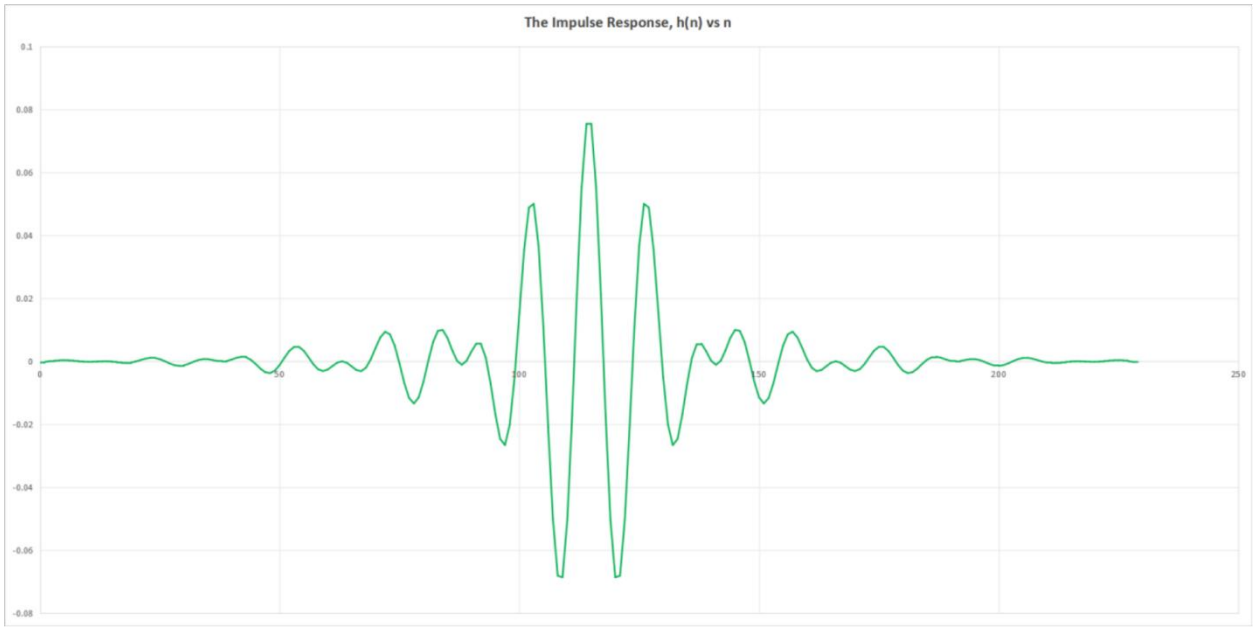
So, need to calculate h(n) for n=0,1, 2, ... ,229. This ensures that the filter can achieve sufficient attenuation and frequency sharpness to effectively filter EEG signals, especially alpha waves (8–13 Hz).

To generate an FIR filter using the window method, a table in Microsoft Excel is arranged with several important columns, namely n, n - α, hD(n), w(n), and h(n). Column n represents the time indices from 0 to 229 because this filter has a length of 230 points (N+1), in line with the maximum allowed order limit of 229. Column n - α shows the distance of each index from the centre point of the filter (α = N/2 = 114.5). This is important because the ideal impulse function, hD(n), depends on the relative value between each n and the centre α, which determines the symmetry of the filter.

Next, column hD(n) calculates the ideal impulse value of the band-pass filter. The formula hD(n) = sin (w2(n - α)) - sin (w1(n - α))/pi (n - α) is used. Column w(n) represents a window function, such as a Hamming window, which is used to smooth the edges of hD(n) and reduce the effects of side frequencies. Finally, h(n) is the result, which is the actual filter coefficient used in the EEG filtering system, obtained by multiplying the values of hD(n) and

w(n) at each n. This combination ensures that the resulting filter is more robust, efficient, and suitable for filtering alpha waves in EEG signals.

6. Then, plot the impulse response, *h*( ).



The figure above shows the impulse response graph h(n) of a band-pass FIR filter designed to filter EEG alpha waves in the frequency range of 8–13 Hz. The h(n) waveform is symmetrical about the midpoint (around n = 114.5), reflecting the linear-phase nature of the FIR filter. The highest amplitude is in the middle and gradually decreases towards both ends, indicating the use of a Hamming window that smooths the ideal impulse function. This property is important for reducing the effect of the "Gibbs phenomenon" and ensuring stable filtering performance in the frequency domain. The Gibbs phenomenon is visible by appearing as small waves that oscillate around the edges of the signal or transition band. It never completely disappears even if add more samples but can be reduced using windowing such as Hamming or Blackman.

This impulse response describes the contribution of each input sample to the output after filtering. Positive and negative values in h(n) reflect the band-pass effect, where only frequencies between f1 and f2 are allowed to pass. The oscillatory pattern around the centre indicates the presence of a frequency component concentrated in the alpha band, and the decay towards the ends indicates a softer, less sharp filter boundary, as required by the Hamming window. Overall, this graph confirms that the filter is properly designed and suitable for EEG applications. And its value of peak magnitude same as calculation (13) which is 0.0781.

TASK 3 (10 marks)

Develop the programming algorithms to solve the problem. Show and explain the frequency response specifications.

7. Design your filter in MATLAB.

Objective: Design a Bandpass FIR filter for alpha waves (8–13 Hz) using sampling frequency = 128 Hz.

Frequency Response Specifications

Filter Type: Bandpass

Cutoff frequencies: 8 Hz and 13 Hz

Sampling rate: 128 Hz

Normalized cutoff: [0.125, 0.2031]

Filter order: 229

Window used: Hamming (provides good stopband attenuation)

Algorithm in MATLAB (Using Window Method)

FIR_Alpha_Filter_Design.m | FIR_AlphaBand_Filter.m | AlphaG8.m

C:\Users\HP\OneDrive\Documents\MATLAB\FIR_Alpha_Filter_Design.m

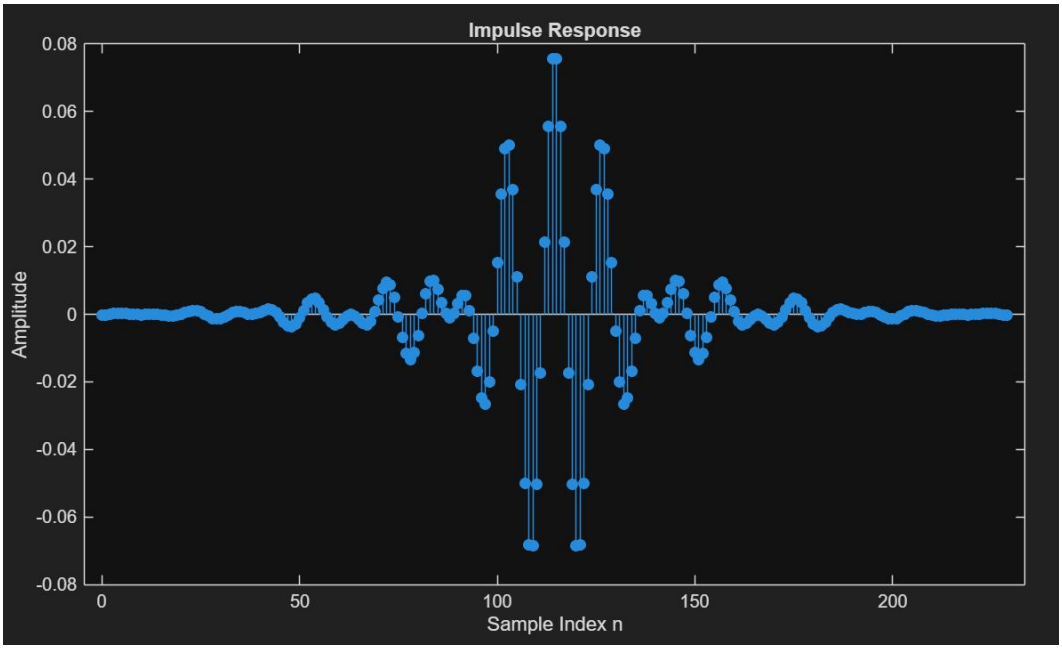
```
% FIR Bandpass Filter for Alpha band (8-13 Hz)
Fs = 128;              % Sampling Frequency in Hz
f1 = 8;                % Lower cutoff frequency in Hz
f2 = 13;               % Upper cutoff frequency in Hz
N = 229;               % Filter order (can go up to 229)
wn = [f1 f2]/(Fs/2);   % Normalize cutoff frequencies (0 to 1)

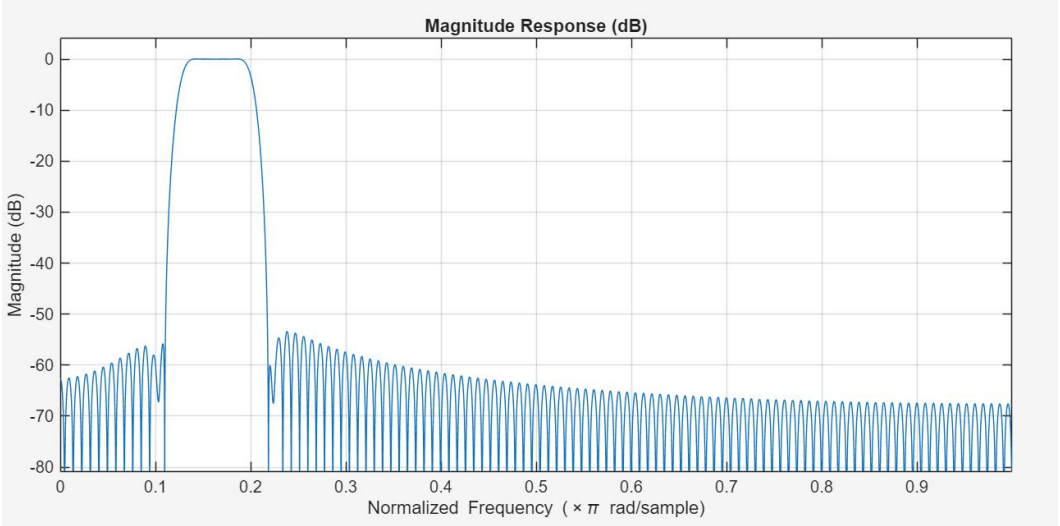
% Design FIR filter using Hamming window
b = fir1(N, wn, 'bandpass', hamming(N+1));

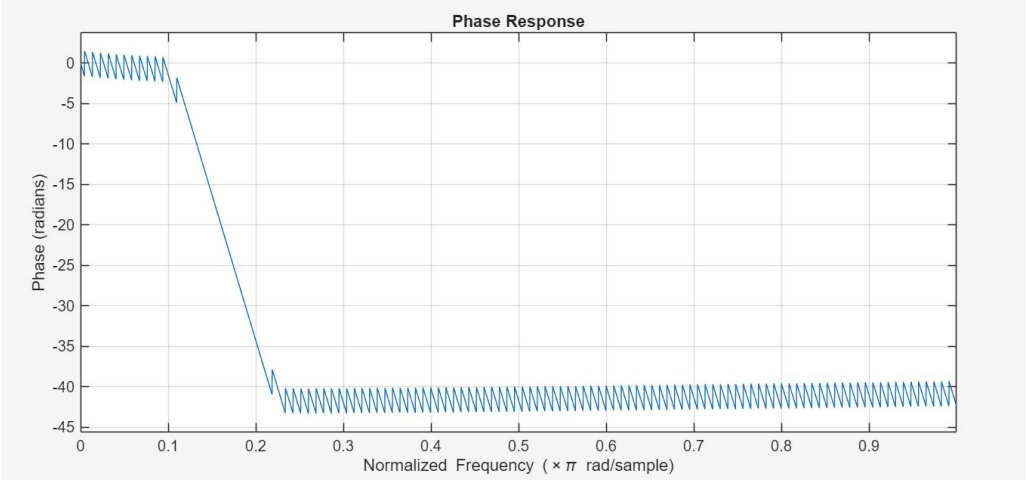
% Plot Impulse Response
figure;
stem(0:N, b, 'filled');
title('Impulse Response');
xlabel('Sample Index n');
ylabel('Amplitude');

% Plot Frequency Response
fvtool(b, 1);  % Interactive magnitude & phase plot
```

For Filter Order, N=229



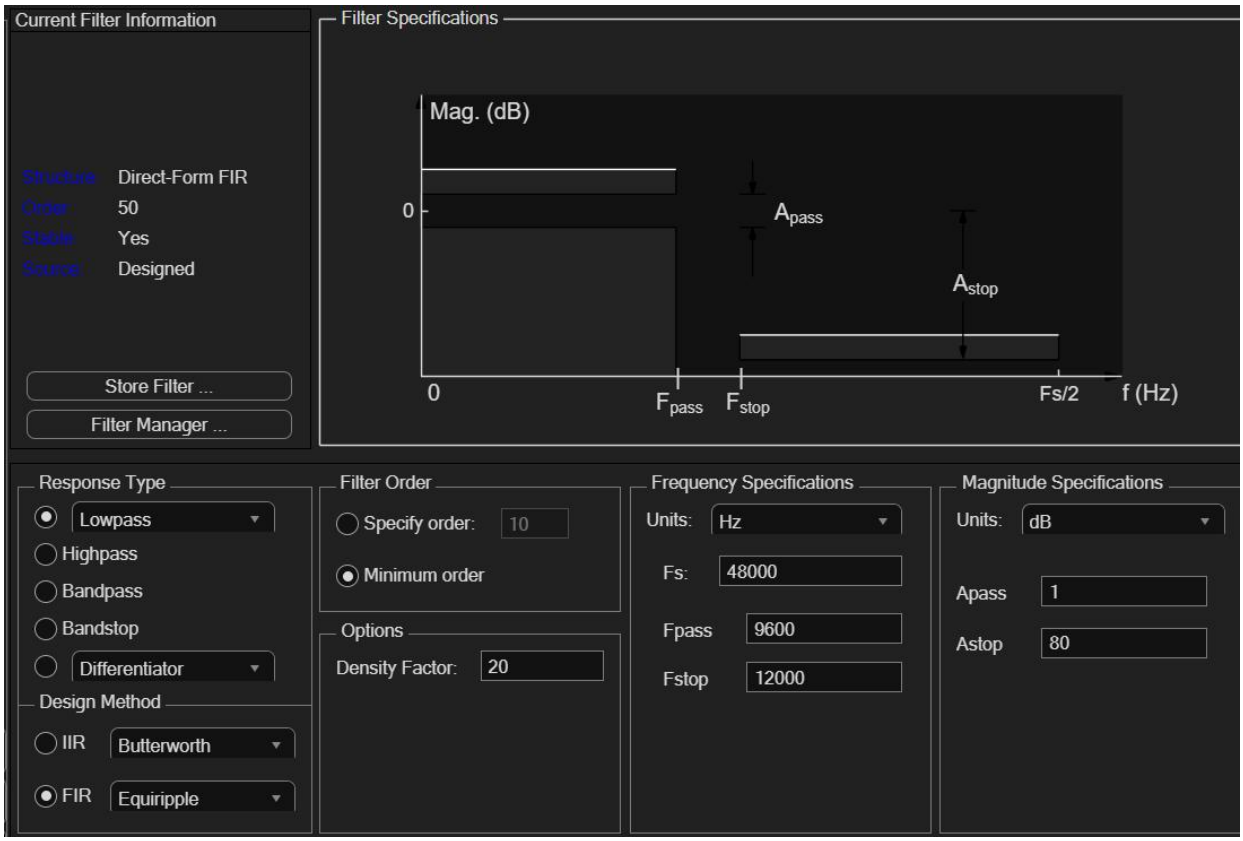




Designing a Bandpass FIR filter to separate alpha waves between 8 and 13 Hz from EEG signals collected at 128 Hz is the aim of this exercise. A bandpass filter type with cutoff frequencies of 8 Hz and 13 Hz, normalized to [0.125, 0.2031] for MATLAB implementation, is part of the frequency response specifications. The Hamming window is used to give good stopband attenuation and minimum ripple in comparison to other windows, and the filter order is set at 229, the greatest permitted, to guarantee a narrow transition band and improved frequency selectivity. The filter design algorithm consists of defining the cutoff points and sample frequency, normalizing them, and then using MATLAB's fir1() function with the given order and window. While the frequency response demonstrates that the filter efficiently passes the alpha band (8–13 Hz) while attenuating other frequencies, such as delta, theta, and beta, the impulse response of the constructed filter validates a linear-phase characteristic. For offline EEG signal processing, using a high-order FIR filter improves accuracy but adds computing overhead, which is acceptable. Through MATLAB simulation and FVTool visualization, the design was verified, ensuring that the filter satisfies the specified requirements.

8. Assess and plot the frequency response and impulse response from filterDesigner.

Step 1 : Open **filterDesigner** in matlab



Step 2:

input the following settings:

Response Type: Bandpass

Design Method: FIR → Window

Order: 229

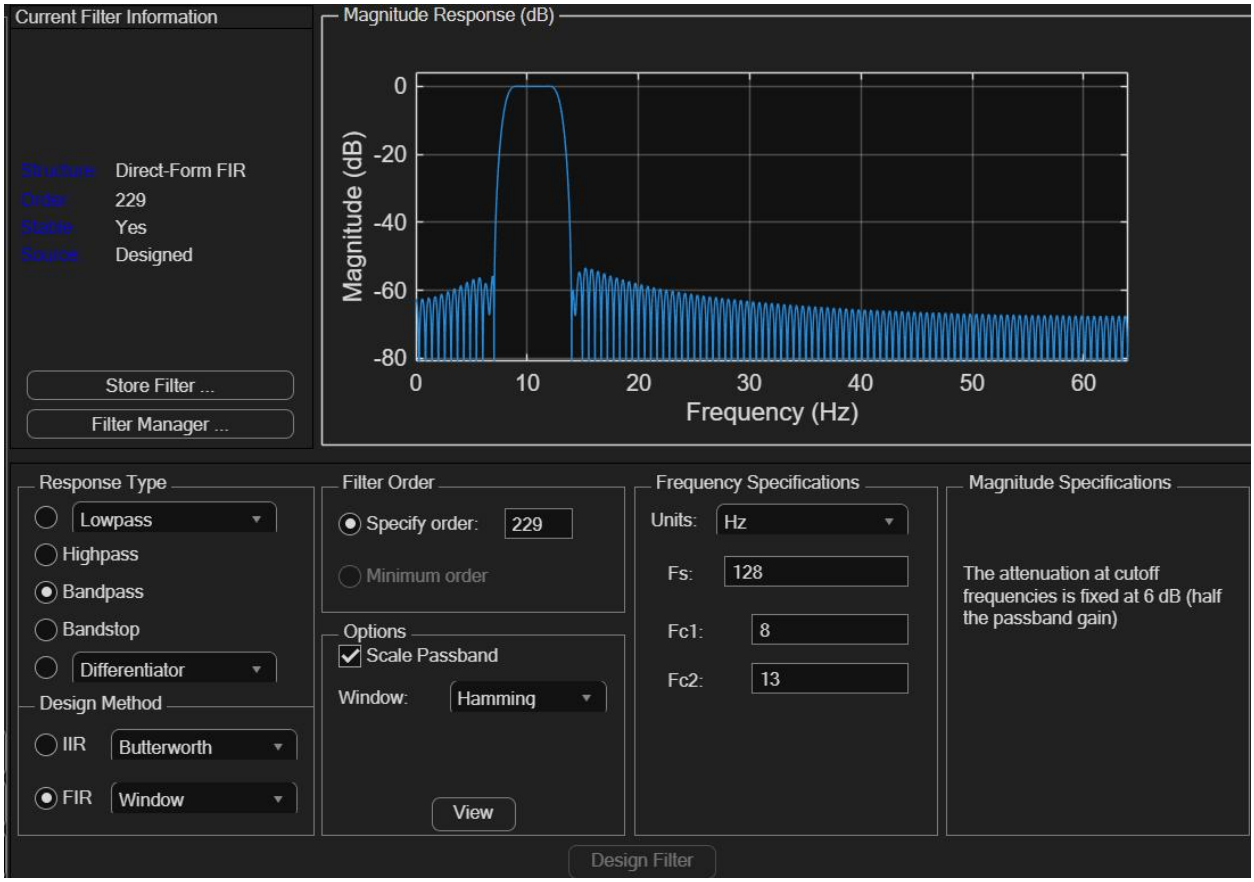
Fs: 128 Hz

Fc1: 8 Hz

Fc2: 13 Hz

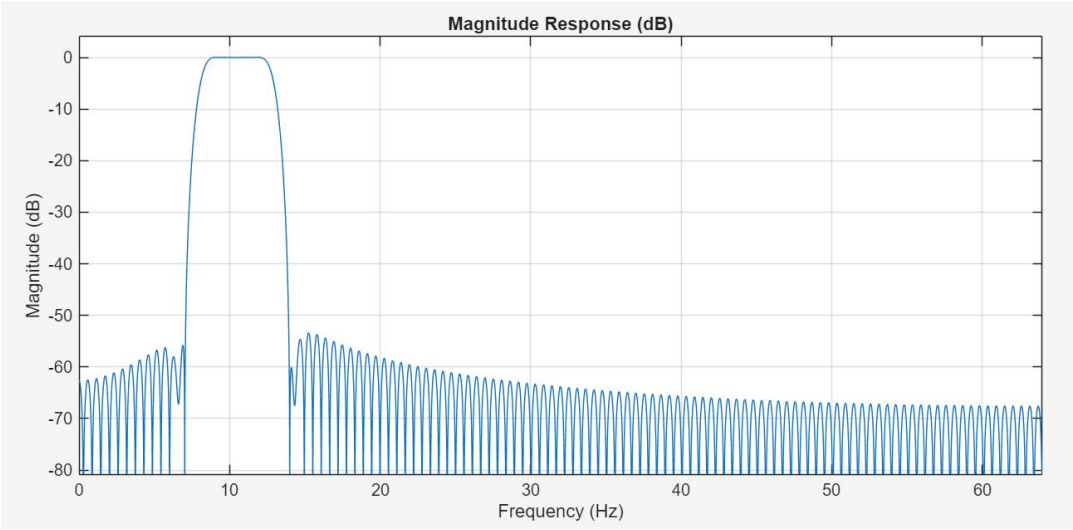
Window: Hamming

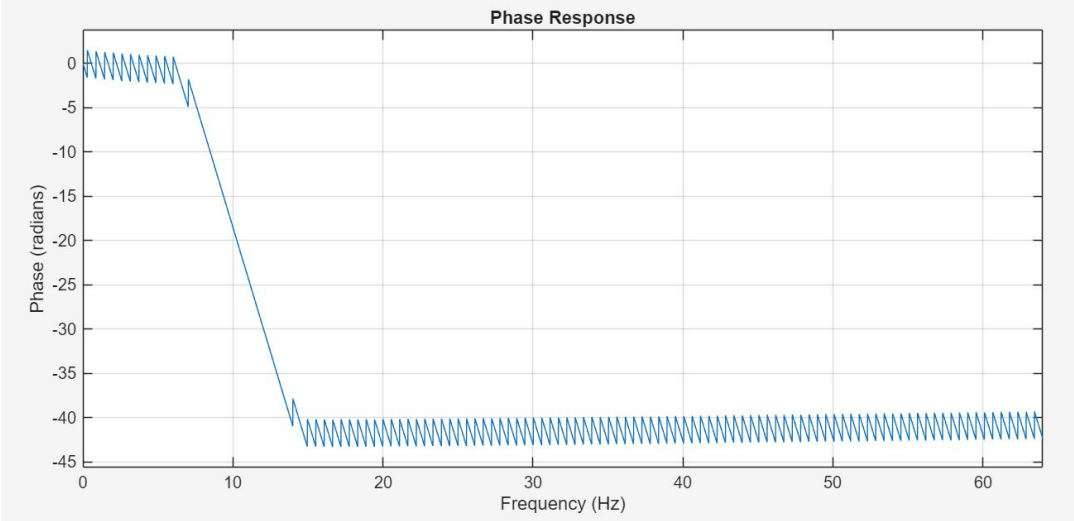
Step 3: Click “Design Filter”

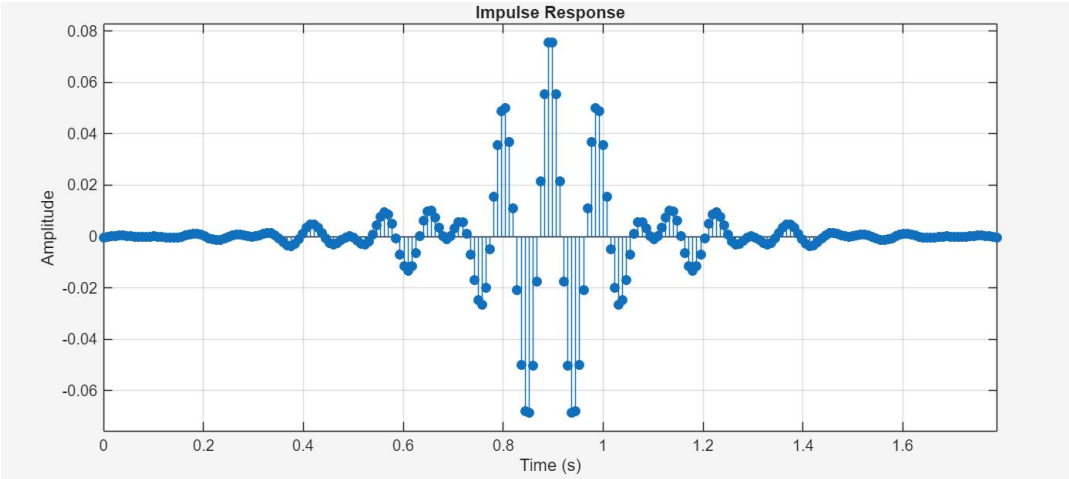


Step 4: Click View → Filter Visualization Tool (FVTool) to display the response

Magnitude response: Shows how well 8–13 Hz is preserved




Phase Response
Phase (radians)
0
-5
-10
-15
-20
-25
-30
-35
-40
-45
0
10
20
30
40
50
60
Frequency (Hz)

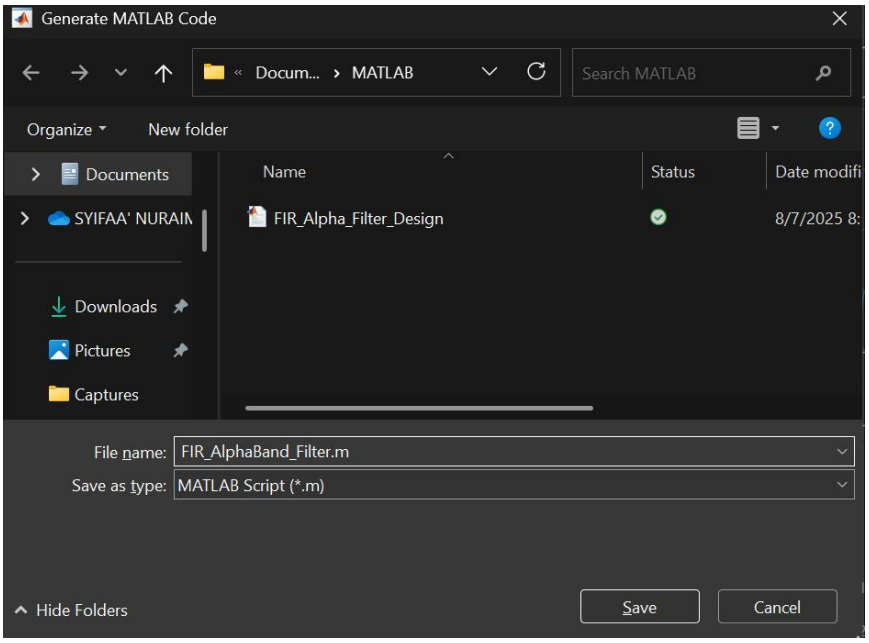

Impulse Response
Amplitude
0.08
0.06
0.04
0.02
0
-0.02
-0.04
-0.06
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
Time (s)

9. Export the m-file for the designed filter.

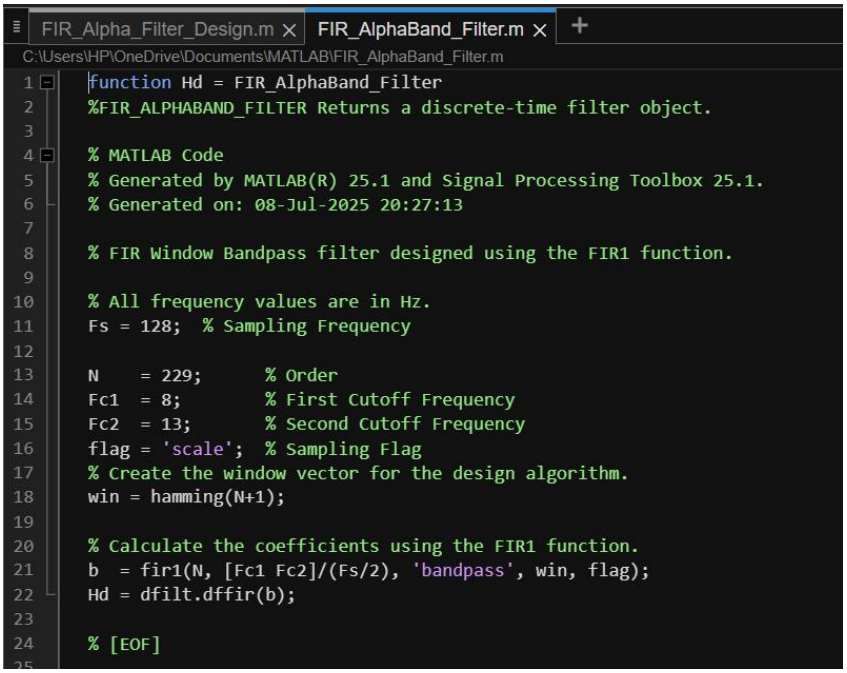
In filterDesigner, go to:

File → Generate MATLAB Code → Filter Design Function

Save the file as : FIR_AlphaBand_Filter.m



MATLAB will auto-generate the code:



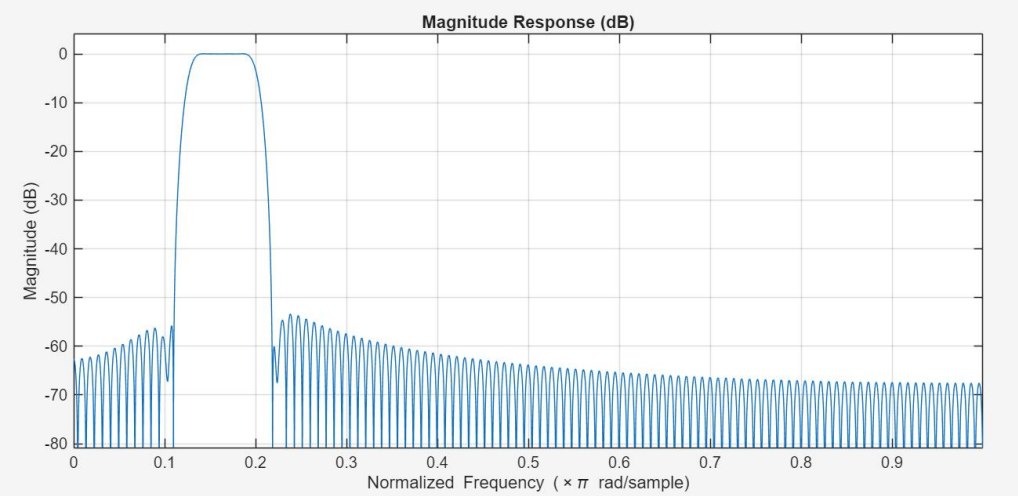
This script uses the Hamming window to recreate the FIR filter, which is then visualized in FVTool.

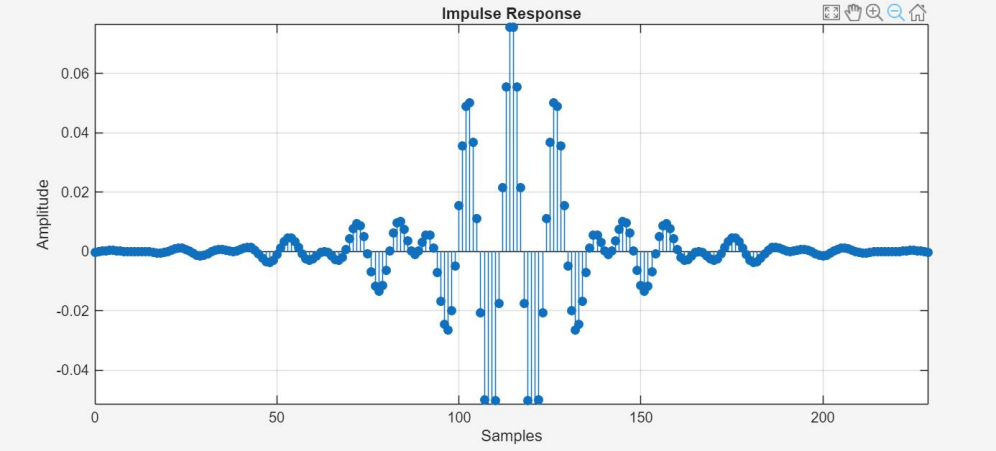
Run exported and converted **.m** function file.

```
Command Window
New to MATLAB? See resources for Getting Started.

>> Hd = FIR_AlphaBand_Filter;
>>  fvtool(Hd);
```

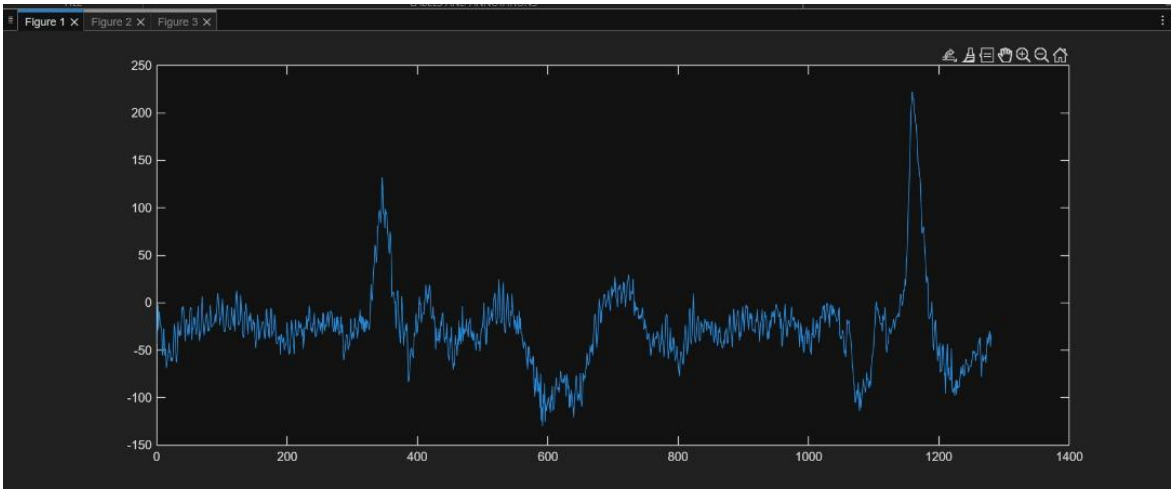
Output:



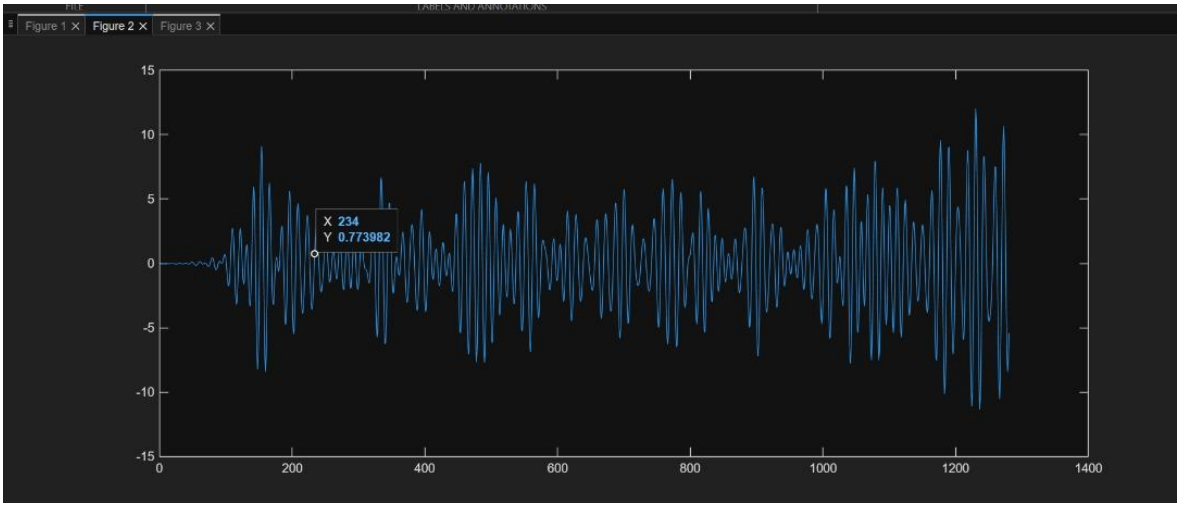


The output question 9 similar to question 7, because the same filter design problem creating a bandpass FIR filter for alpha waves (8–13 Hz) with the same parameters (Fs = 128 Hz, order = 229, Hamming window) is being addressed by both Questions 7 and 9. Therefore, it stands to reason that the filter responses will appear nearly identical. Indeed, since the filter was created using two distinct techniques, the output looks the same. This demonstrates consistency, the filter is appropriately built whether code (Q7) or GUI (Q9) is used. Thus, this confirms the accuracy of both manual and GUI-based design.

TASK 4 (5 Marks)



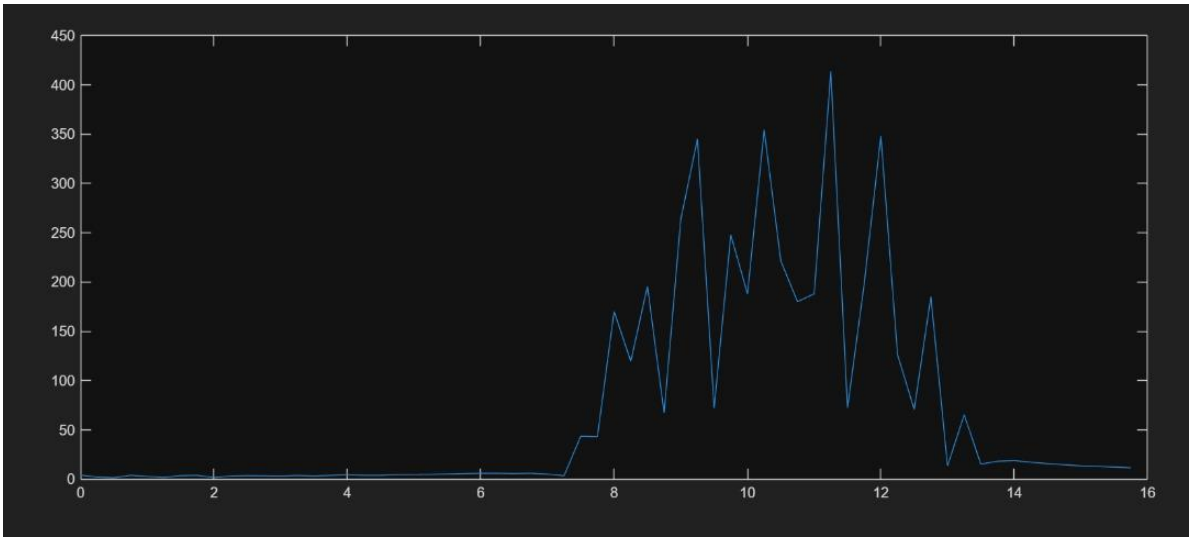
The bandpass FIR filter designed using the hamming window method successfully isolated the alpha wave frequency band (8–13 Hz) from the EEG signal. Designed with a sampling rate of 128 Hz, filter order of 229, and a Hamming window, the filter aimed to balance computational efficiency with effective frequency selectivity.



The frequency spectrum obtained from MATLAB simulation confirms that the filter performs according to the specifications. The signal energy is well-concentrated in the desired 8–13 Hz passband, with a significant drop outside this range, indicating that both lower (delta, theta) and higher frequency components (beta, noise) are attenuated.

However, minor signal leakage is observed just before 8 Hz and just above 13 Hz. This is expected, as the Hamming window provides a typical stopband attenuation of approximately –53 dB, and its transition band is not as sharp as that of more aggressive windows. The

observed leakage occurs in the transition regions where the filter rolls off gradually rather than abruptly.



To further improve the performance, especially to reduce the leakage outside the passband, the following enhancements could be considered such as by increasing the filter order, which narrows the transition band. Moreover, switching to a window with better stopband attenuation, such as the Blackman or Kaiser window and using an optimal FIR design approach, such as the equiripple (Parks-McClellan) algorithm, which allows precise control over passband and stopband ripple and sharper cutoffs. The filter formulation was derived mathematically using the window method and implemented in both MATLAB and Microsoft Excel. In Excel, filter coefficients were manually computed based on the theoretical impulse response equation.

Although there are no sophisticated signal processing visualization capabilities in Excel, this step was helpful in understanding how each coefficient shaped the filter's response. Both manual coding and the filterDesigner GUI's final implementation in MATLAB verified that the filter satisfies the theoretical requirements of a FIR design based on a Hamming window. The filter's bandpass nature is evident in the FVTool visualization, which also displays a well-defined passband and anticipated sidelobe behavior that is in line with the settings and method used.

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